

Nourhan Bayasi, Ph.D.

📞 +1 604 723 9473 | ✉️ nourhanbayasi92@gmail.com | 🏠 Vancouver, Canada

🌐 LinkedIn | 🎓 Google Scholar | 🐙 GitHub | 🌐 Website

About Me

I'm an AI scientist and postdoctoral researcher developing physician-in-the-loop and agentic AI models for medical imaging, with a focus on solutions that translate into real clinical practice. My background spans both academia and industry, giving me a balance of rigor and practicality. I'm passionate about collaborative work and actively mentor junior researchers, helping them build strong, critical thinking skills.

Professional Experience

Postdoctoral Research Fellow

BC Cancer | UBC Faculty of Medicine

Vancouver, Canada

07/2025 - present

- Developing physician-in-the-loop and agentic AI systems for CT and PET workflows.
- Designing interactive AI systems integrating clinician feedback to improve robustness and deployment readiness.
- Translating research into clinically applicable AI solutions.

Founder and Mentor

Rising Scholars Hub (RiSH)

Online

03/2025 - present

- Founded a global mentorship initiative supporting early-career researchers in AI and healthcare.
- Organized workshops and collaborative research activities.
- Built an international research network across disciplines.

Graduate Teaching Assistant

University of British Columbia (UBC)

Vancouver, Canada

09/2020 - 04/2025

- Delivered instruction across labs, tutorials, and office hours.
- Evaluated assignments, exams, and projects with consistent academic standards.
- Coordinated lab logistics and setup.

Machine Learning Engineer Intern

Cognia AI

Vancouver, Canada

04/2024 - 10/2024

- Developed bias mitigation algorithms to improve fairness in ML systems.
- Enhanced continual learning pipelines for evolving data.
- Contributed to trustworthy AI development practices.

Graduate Academic Assistant

University of British Columbia (UBC)

Vancouver, Canada

06/2020 - 08/2020

- Designed online course materials and assessments.
- Integrated digital tools to improve engagement.

Lab Instructor, Electrical Engineering

Higher Colleges of Technology (HCT)

Sharjah, UAE

08/2017 - 12/2019

- Delivered lab instruction and supervised student projects.
- Managed lab operations and safety systems.
- Mentored students to six top-three national competition wins.

Workshop Engineer, Electrical Engineering

Institute of Applied Technology (IAT)

Umm Al Quwain, UAE

09/2015 - 07/2017

- Supported curriculum delivery and student projects.
- Mentored students to multiple first-place science fair wins.
- Managed lab setup, maintenance, and procurement.

Academic Education

PhD in Electrical and Computer Engineering

University of British Columbia (UBC)

Vancouver, Canada

01/2020 – 03/2025

- Thesis: *Beyond Catastrophic Forgetting: Advancing Continual Learning for Robust and Fair Medical Image Analysis*
- CGPA: 94.1%

Master of Science in Electrical and Computer Engineering

Khalifa University (KU)

Abu Dhabi, United Arab Emirates

09/2013 – 04/2015

- Thesis: *An ECG signal processor for the Prediction of ventricular arrhythmia*
- CGPA: 3.7/4

Bachelor of Science in Communication Engineering

Khalifa University (KU)

- CGPA: 3.96/4 (1st in cohort)

Sharjah, United Arab Emirates

09/2009 – 04/2013

Technical Skills

Languages

Python, MATLAB, C++

Frameworks

PyTorch, Keras, TensorFlow, Scikit-learn, OpenCV, Hugging Face (Transformers, PEFT), MONAI, SAM/SAM2

Tools/Platforms

Linux, VS Code, Git, Docker, AWS, SLURM/HPC, Weights & Biases, Overleaf

Honors and Awards

Research Awards and Recognition

- **Shortlisted, NSERC and L'Oréal-UNESCO For Women in Science Supplement.** *Invited as the top-ranked woman candidate from an NSERC Canada Postdoctoral Research Award Review Committee.* Canada || 2026
- **3rd Place, Best Paper Award, EMERGE Workshop @ MICCAI** South Korea || 2025
- **1st Place, WiM Best Health Equity Paper Award, MICCAI Conference** Morocco || 2024
- **2nd Place, WiM Best Oral Presentation Award, MICCAI Conference** Morocco || 2024
- **Shortlisted, MICCAI Best Paper Award, MICCAI Conference** Morocco || 2024
- **Shortlisted, MICCAI Young Scientist Award, MICCAI Conference** Morocco || 2024
- **Best Paper Award, ISIC Medical Image Analysis Workshop @ MICCAI** Canada || 2023
- **Best Paper Award, ISIC Medical Image Analysis Workshop @ ECCV** Tel Aviv || 2022
- **Best Paper Award, IEEE Transactions on Very Large-Scale Integration Systems** USA || 2016

Fellowships and Scholarships

- **Canada Postdoctoral Research Award (CPRA), NSERC** Canada || 2026-2028
- **Vanier Canada Graduate Scholarship (Ranked Top 1 Nationwide), NSERC** Canada || 2022-2025
- **Borealis AI Global Fellowship, RBC's AI Research Institute** Canada || 2024
- **Four Year Doctoral Fellowship (4YF), University of British Columbia (UBC)** Canada || 2022-2026
- **International Student Award, University of British Columbia (UBC)** Canada || 2020-2022
- **Scholarship Award, PhD Studies, University of British Columbia (UBC)** Canada || 2020-2023
- **Scholarship Award, Master Studies, Khalifa University (KU)** UAE || 2013-2015
- **Scholarship Award, Bachelor Studies, Khalifa University (KU)** UAE || 2009-2013

Academic and Institutional Awards

- **Dr. and Mrs. Brandwajn Graduate Award, University of British Columbia (UBC)** Canada || 2021
- **Faculty of Applied Science Graduate Award, University of British Columbia (UBC)** Canada || 2021
- **President's Academic Excellence Initiative PhD Award, University of British Columbia (UBC)** Canada || 2021
- **Society Registration Grant, MICCAI Conference** Morocco || 2024
- **Student Registration Grant, MICCAI Conference** France || 2021
- **Best Prototype Award, 2nd Place, Engineering Student Renewable Energy Competition @ UAE University** UAE || 2013
- **Best Poster Award, 1st Place, Undergraduate Research Conference on Applied Computing @ Zayed University** UAE || 2013
- **Leadership Award, Best Student Category, Khalifa University (KU)** UAE || 2013
- **President List Award, College of Engineering, Khalifa University (KU)** UAE || 2009-2013

Leadership and Service

- **Founder and Organizer, CLiMeM Workshop @ MICCAI** 2026 – Current
- **Judge, AI and Machine Learning Session, UBC Radiology Clinical & Fundamental Science Research Day** 2026
- **Public Outreach and Membership Officer, Women in MICCAI (WiM)** 2025 – Current
- **Doctoral Programs Officer, MICCAI Student Board** 2025 – Current
- **Electronics Expert, Emirates Skills National Competition, UAE** 2017
- **Standard Leader, AdvancED Academic Accreditation, Institute of Applied Technology, UAE** 2016
- **STEAM Program Developer, Engineering, Institute of Applied Technology, UAE** 2016

Professional Activities

- Reviewer, *MICCAI, ISIC Skin Image Analysis Workshop @ MICCAI* 2022 – Current
- Reviewer, *ECCV, NeurIPS, IEEE TMI* 2026
- Reviewer & Emergency Reviewer, *MICCAI, WACV, MSB EMERGE @ MICCAI, MedIA, IEEE TNNLS (A/E)* 2025
- Reviewer, *CVPR, Medical Image Analysis (MedIA), IEEE TMI* 2024

Personal Information

- Nationality: *Canadian*
- Date of Birth: *15 January 1992*